# 1. Python SQLite Game Shop POS

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Time required: 60 minutes

- Follow all directions carefully and accurately.
- Think of the directions as minimum requirements.

## SQL Tutorial

- https://www.w3schools.com/sql/sql_create_table.asp

## SQLite with Python Tutorials

- SQLite Databases with Python - Full Course – FreeCodeCamp.org
- https://www.sqlitetutorial.net

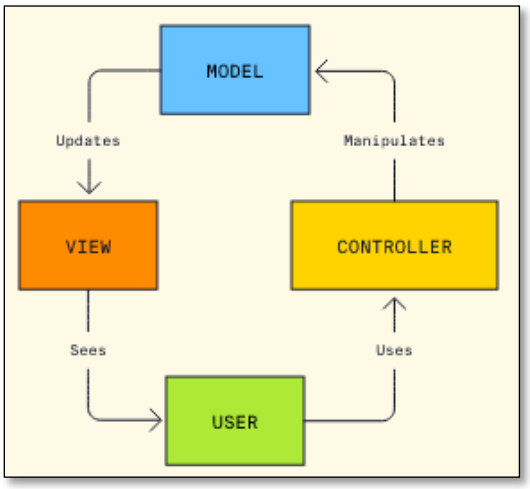
## MVC (Model View Controller)

MVC is short for Model, View, and Controller. MVC is a popular way of organizing your code. The big idea behind MVC is that each section of your code has a purpose, and those purposes are different. Some of your code holds the data of your app, some of your code makes your app look nice, and some of your code controls how your app functions.

**Model:** Model code typically reflects real-world things. This code can hold raw data, or it will define the essential components of your app. For instance, if you were building a To-do app, the model code would define what a "task" is and what a "list" is – since those are the main components of a todo app.

**View:** View code is made up of all the functions that directly interact with the user. This is the code that makes your app look nice, and otherwise defines how your user sees and interacts with it.

**Controller:** Controller code acts as a liaison between the Model and the View, receiving user input and deciding what to do with it. It's the brains of the application, and ties together the model and the view.




## SQLite Relational Database

SQLite is a relational database. We can create tables related by primary keys. We will design our databases using an ERD (Entity Relationship Diagram). www.lucidchart.com is free web-based diagram site used in these SQLite tutorials.

In this tutorial, we will create a games database with two related tables, game and publisher along with a bridge or transaction table.

### Business Rules

Business rules establish how the entities interact and their relationship with each other. For example:

- A publisher can produce many games.
- A game can only have one publisher.

- A customer can purchase many games.
- A game can be purchased by many customers.

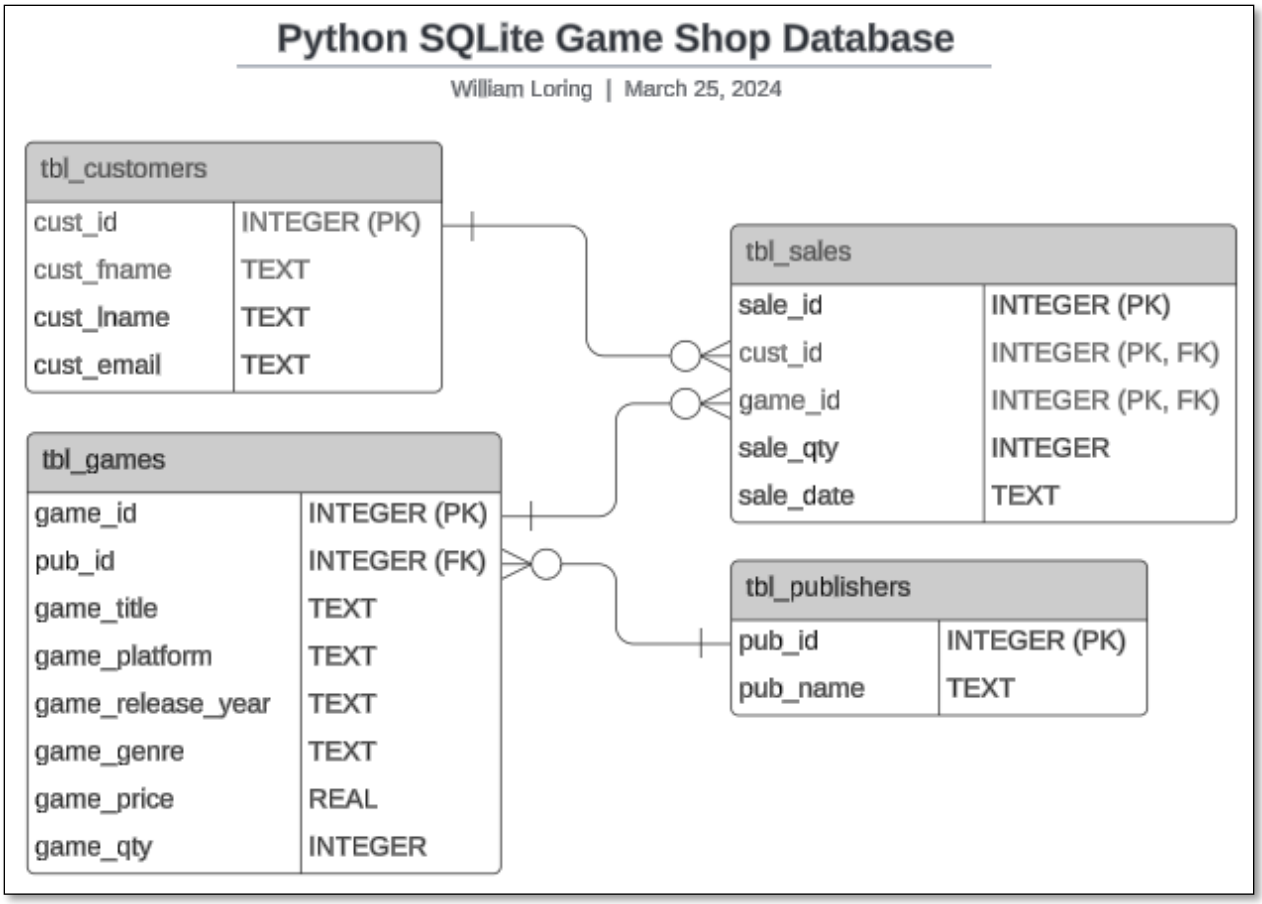
## Game Shop Database ERD

An Entity Relationship Diagram, also known as ERD, is a diagram that displays the relationship of entities stored in a database. ER diagrams help to explain the logical structure of databases. ER diagrams are created based on three basic concepts: entities (tables), attributes (fields), and relationships.

An ERD of our movies related database is based on the following business rules:

- A publisher can produce many games.
- A game can only have one publisher.
- A customer can purchase many games.
- A game can be purchased by many customers.

This is a crow’s foot diagram illustrating a one to many relationship between the entities.

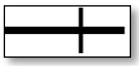



Reference: [https://vertabelo.com/blog/crow-s-foot-notation/](https://vertabelo.com/blog/crow-s-foot-notation/)

**PK (Primary Key):** A primary key is a column or a set of columns in a table whose values uniquely identify a row in the table. A primary key typically has no meaning other than to uniquely identify each record.

**FK (Foreign Key):** A foreign key is a column or a set of columns in a table whose values correspond to the values of the primary key in another table.

Crows foot notation indicates the record relationship between the tables.

| | |
|---|---|
| | One or many |
|  | One |

## Assignment 1: Draw ERD

1. Take your ERD from your previous database project.
2. Draw a Crow's Foot ERD for a relational table version of your project similar to the example.
3. Create a Word Documentation file to store a screenshot of your ERD and your other planning items.

## Assignment 2: Data Dictionary

Let's create your database and tables. Fill in your names in your planning document.

1. Create a Python file named:
2. Database file name:
3. Tables (Entities):
4. Fields (Attributes)

This is the structure of our sample database. This is called a data dictionary.

| Table Name | Field Name | Field Format | Description |
|---|---|---|---|
| tbl_publishers | pub_id | INTEGER – PK | Game publisher unique identifier |
| tbl_publishers | pub_name | TEXT | Game publisher name |
| tbl_games | game_id | INTEGER – PK | Game unique identifier |
| tbl_games | pub_id | INTEGER – FK | Foreign key referencing the publisher associated with the game |
| tbl_games | game_title | TEXT | Game title |
| tbl_games | game_platform | TEXT | Game platform |
| tbl_games | game_release_year | INTEGER | Game release year |
| tbl_games | game_genre | TEXT | Game genre |
| tbl_games | game_price | REAL | Price of game |
| tbl_games | game_qty | INTEGER | Quantity of games on hand |
| tbl_sales | sale_id | INTEGER (PK) | Sale unique identifier |
| tbl_sales | sale_qty | INTEGER | Sale quantity |
| tbl_sales | sale_date | TEXT | Date of sale |

| | | | |
|---|---|---|---|
| tbl_sales | cust_id | (PK, FK) | Composite key for sales and customers |
| tbl_sales | game_id | (PK, FK) | Composite key for sales and games |
| tbl_customers | cust_id | INTEGER - PK | Customer unique identifier |
| tbl_customers | cust_f_name | TEXT | First name |
| tbl_customers | cust_l_name | TEXT | Last name |
| tbl_customers | cust_email | TEXT | Email Address |

**Assignment:**

1. Update your data dictionary for your database in your planning document.
2. Add a Customer Email address.

## Assignment Submission

1. Attach the program files.
2. Attach screenshots showing the successful operation of the program.
3. Submit in Blackboard.